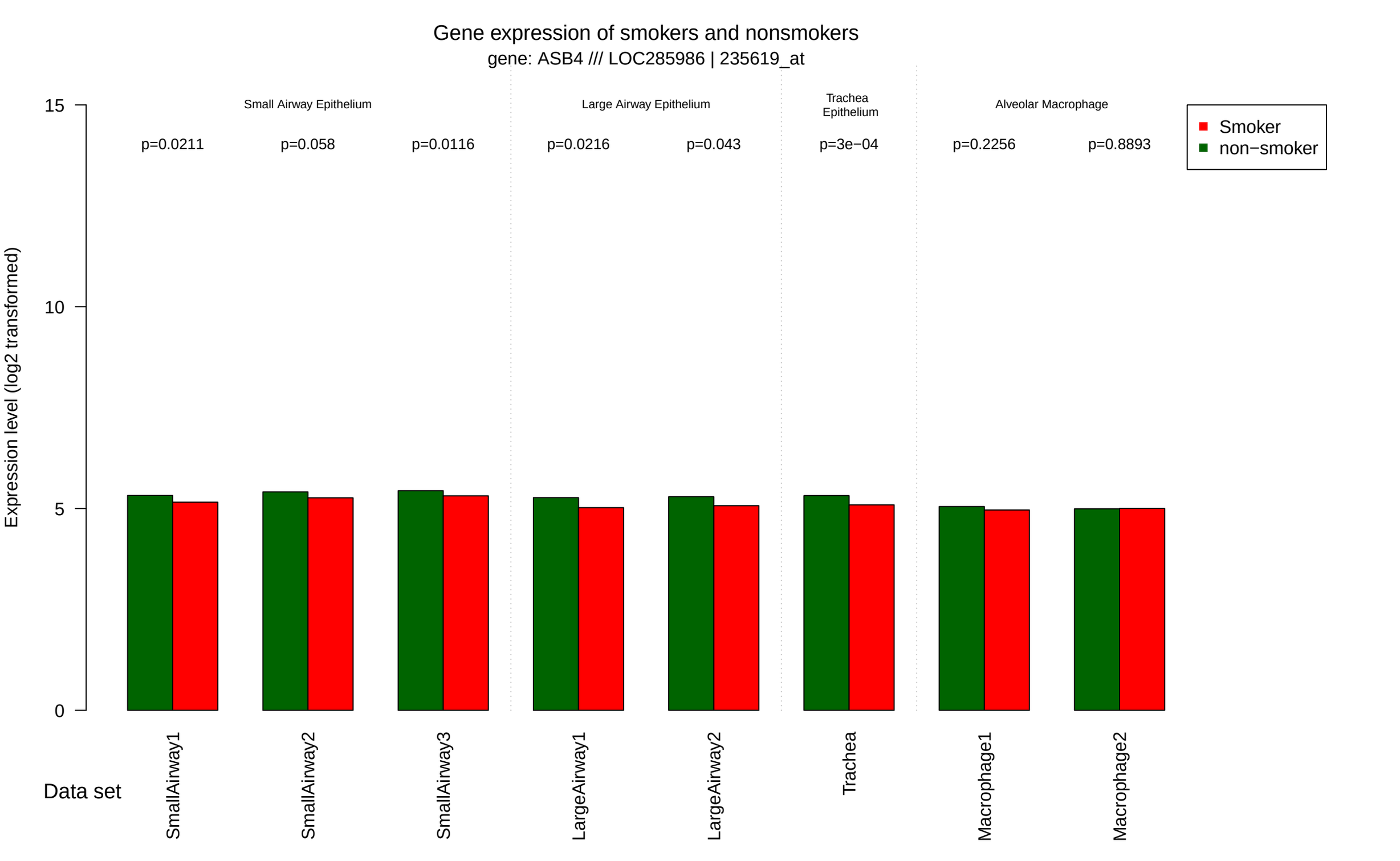
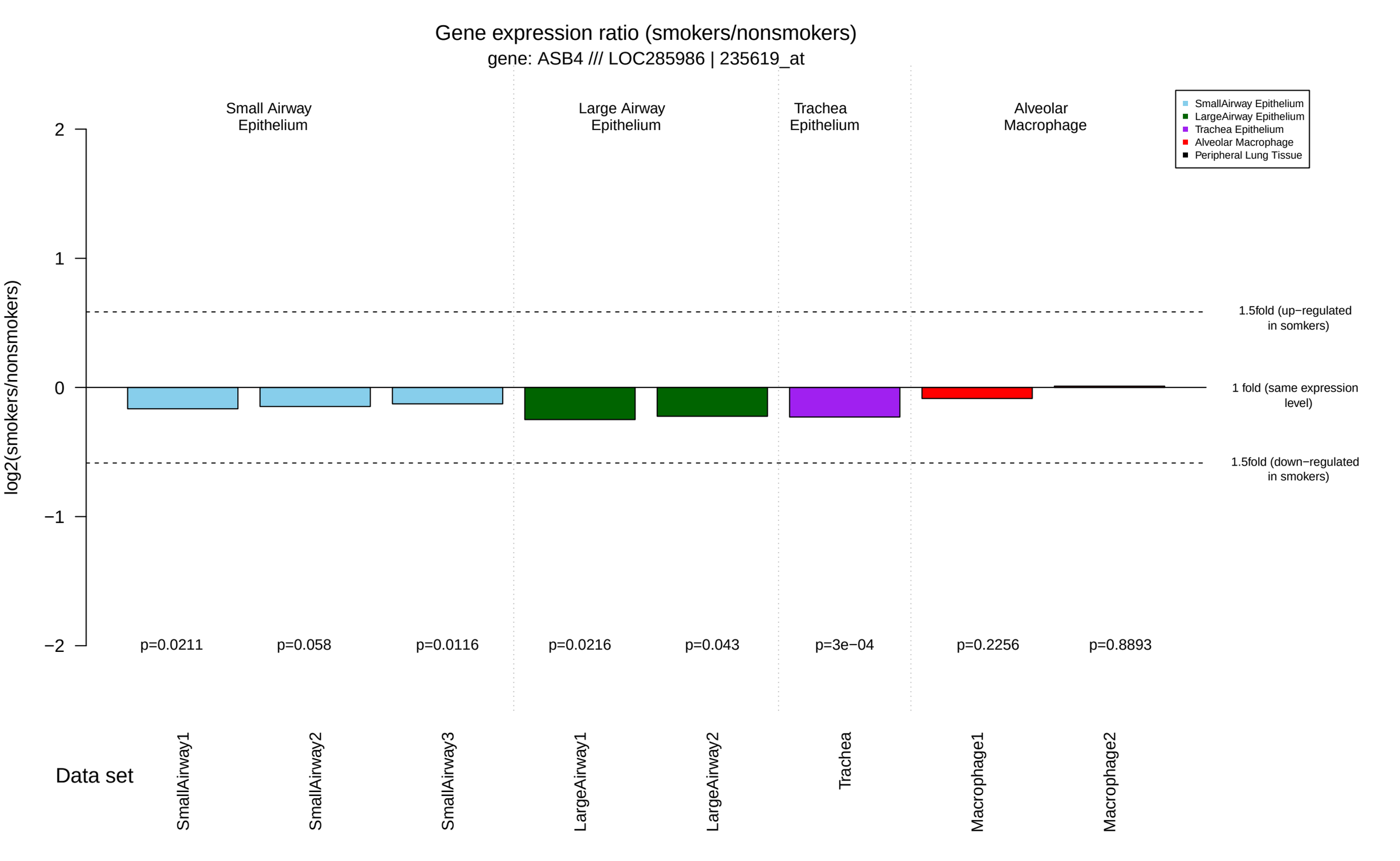
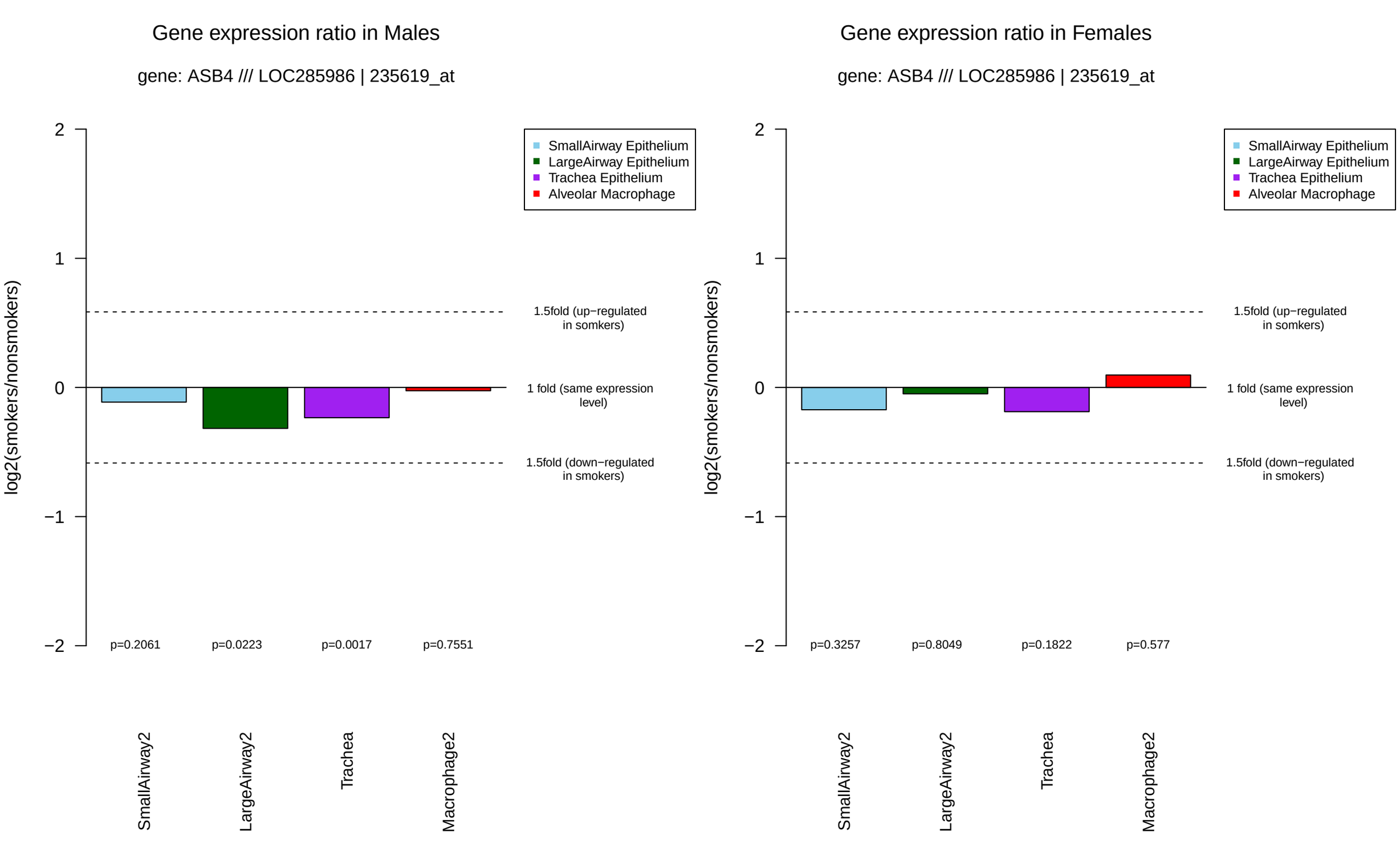
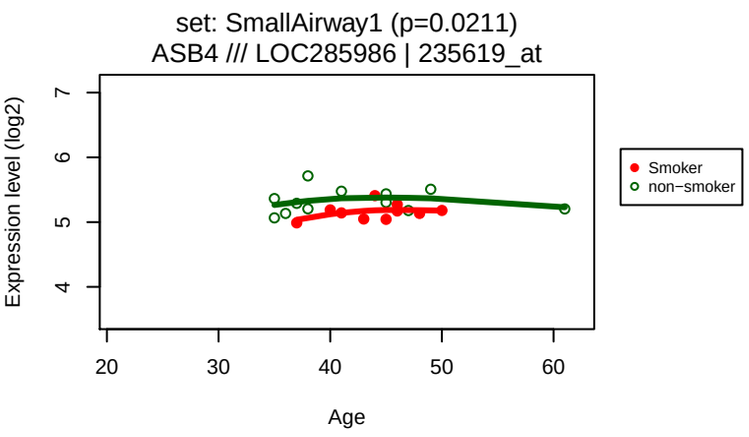
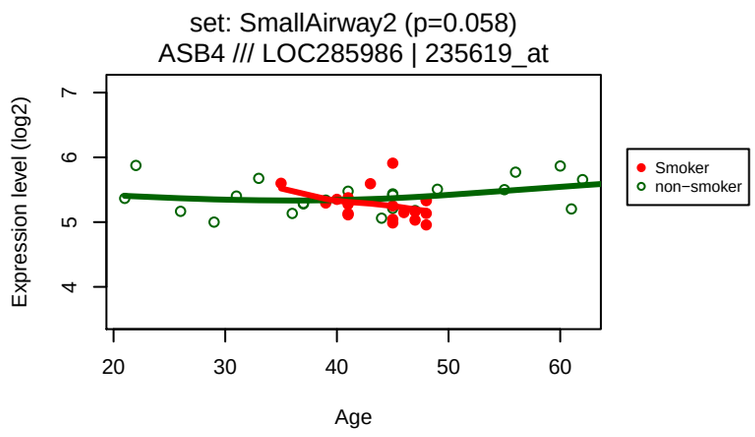
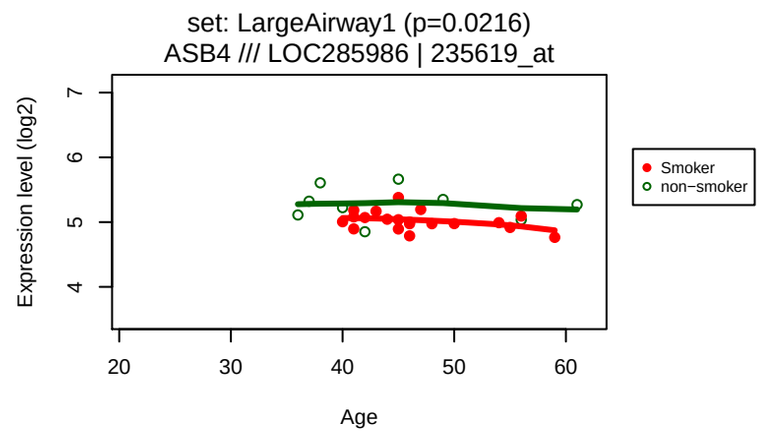

Gene expression of smokers and nonsmokers
gene: ASB4 /// LOC285986 | 235619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0211
p=0.058
p=0.0116
p=0.0216
p=0.043
p=3e−04
p=0.2256
p=0.8893
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

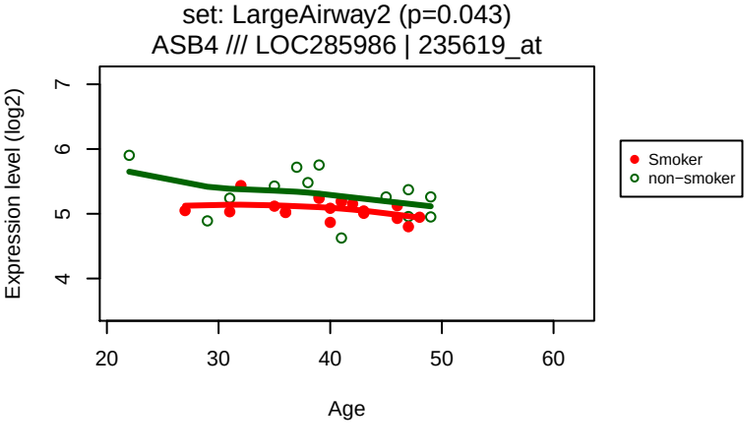

Gene expression ratio (smokers/nonsmokers)
gene: ASB4 /// LOC285986 | 235619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0211
p=0.058
p=0.0116
p=0.0216
p=0.043
p=3e−04
p=0.2256
p=0.8893
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

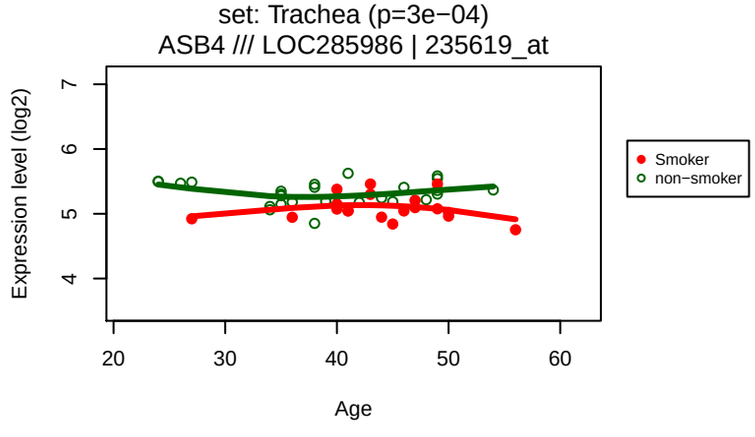

Gene expression ratio in Males
gene: ASB4 /// LOC285986 | 235619_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2061
p=0.0223
p=0.0017
p=0.7551
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASB4 /// LOC285986 | 235619_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3257
p=0.8049
p=0.1822
p=0.577
SmallAirway2
LargeAirway2
Trachea
Macrophage2

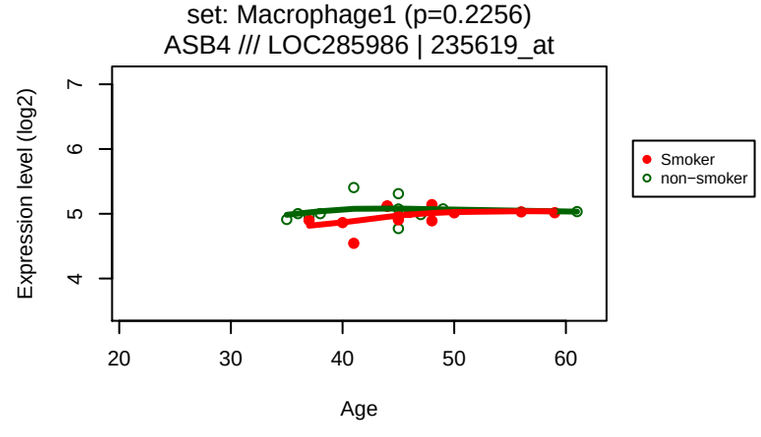

set: SmallAirway1 (p=0.0211)
ASB4 /// LOC285986 | 235619_at
Expression level (log2)
Age
Smoker
non−smoker

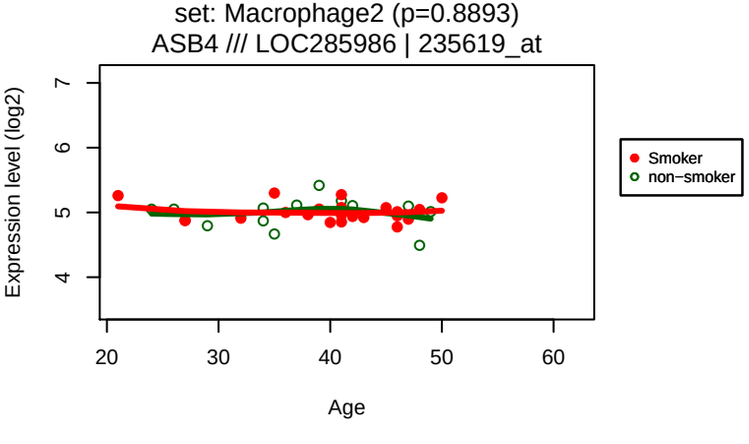

set: SmallAirway2 (p=0.058)
ASB4 /// LOC285986 | 235619_at
Expression level (log2)
Age
Smoker
non−smoker

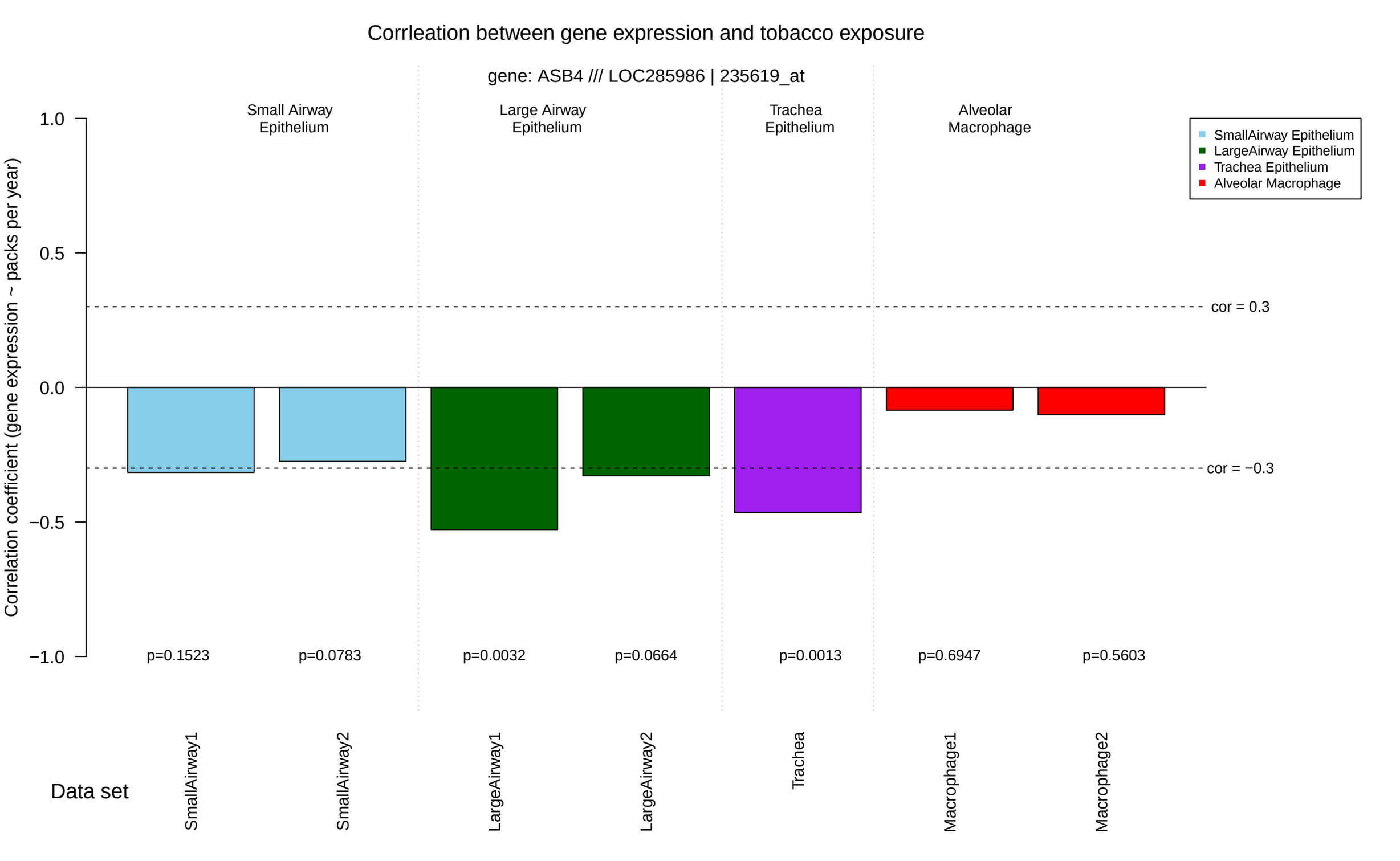

set: LargeAirway1 (p=0.0216)
ASB4 /// LOC285986 | 235619_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.043)
ASB4 /// LOC285986 | 235619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
ASB4 /// LOC285986 | 235619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2256)
ASB4 /// LOC285986 | 235619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8893)
ASB4 /// LOC285986 | 235619_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASB4 /// LOC285986 | 235619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1523
p=0.0783
p=0.0032
p=0.0664
p=0.0013
p=0.6947
p=0.5603
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2